Skills Summary

Convergence Tests for Infinite Series

Use this reference while completing your worksheet or when studying.

Skill 1 — The Ratio Test

When to reach for it: the terms contain $n!$, r^n, or a power $(x-c)^n$ — anything that simplifies when you divide consecutive terms.

Test: let $L = \lim_{n\to\infty} |a_{n+1}/a_n|$. Then $L < 1$ converges absolutely; $L > 1$ (or $L = \infty$) diverges; $L = 1$ tells you nothing — use another test.

Habit: write a_{n+1} and multiply by the RECIPROCAL of a_n; cancel before taking the limit.

Example: Test $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{3^n}$.

Step 1. The term has 3^n in it, so consecutive terms differ by a constant factor — that is exactly what the ratio test exploits.

Step 2. $\displaystyle\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)^2}{3^{n+1}} \cdot \frac{3^n}{n^2} = \frac{1}{3}\left(\frac{n+1}{n}\right)^2 \to \frac{1}{3}.$

$L = \mathbf{\frac{1}{3}} < 1$, so the series converges absolutely.

Try it: Find L for $\displaystyle\sum_{n=1}^{\infty} \frac{n}{4^n}$.

check: $L = \frac{1}{4}$

Skill 2 — The Integral Test

When to reach for it: $a_n = f(n)$ for a function you can actually antidifferentiate — typically $\frac{1}{n^p}$ or something with $\ln n$.

Hypotheses (state them): on $[k, \infty)$, f must be positive, continuous, and decreasing.

Test: $\sum f(n)$ and $\int_k^\infty f(x)\,dx$ converge together. The integral's value is NOT the sum.

Example: Test $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$.

Step 1. The term is $f(n)$ for $f(x) = x^{-3/2}$, which is easy to antidifferentiate, so the integral test is cheaper than any comparison.

Step 2. f is positive, continuous, decreasing on $[1, \infty)$, and $\int_1^\infty x^{-3/2}dx = \lim_{b\to\infty} \left[-2x^{-1/2}\right]_1^b$.

The integral equals **2**, a finite number, so the series converges.

Try it: Evaluate $\displaystyle\int_1^\infty \frac{dx}{x^3}$ and say what it proves about $\sum 1/n^3$.

check: $\mathbf{\frac{1}{2}}$, so the series converges

Skill 3 — Direct Comparison

When to reach for it: the term is "almost" a geometric series or a p-series, spoiled by one extra piece.

Test (non-negative terms only): if $0 \le a_n \le b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges. If $a_n \ge c_n \ge 0$ and $\sum c_n$ diverges, then $\sum a_n$ diverges.

Benchmarks: $\sum r^n$ converges iff $|r| < 1$; $\sum 1/n^p$ converges iff $p > 1$.

Example: Test $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2^n + n}$.

Step 1. Adding n to the denominator only makes each term smaller, so compare upward against the geometric series it came from.

Step 2. $0 < \dfrac{1}{2^n + n} \le \dfrac{1}{2^n}$, and $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2^n}$ is geometric with $r = \frac{1}{2}$, summing to $\dfrac{1/2}{1 - 1/2}$.

The benchmark sum is **1**, finite, so the original series converges.

Try it: Which benchmark bounds $\displaystyle\sum_{n=1}^{\infty} \frac{1}{3^n + 1}$, and what is its sum?

check: $\sum 1/3^n = \mathbf{\frac{1}{2}}$

Skill 4 — Alternating Series and the Error Bound

Test: $\sum (-1)^{n+1} b_n$ with $b_n > 0$ converges when b_n is decreasing AND $b_n \to 0$. Both conditions must be checked.

Error bound: $|S - S_N| \le b_{N+1}$ — the FIRST term you left out, never the last one you kept.

Conditional vs absolute: if $\sum b_n$ also converges the convergence is absolute; if $\sum b_n$ diverges it is only conditional.

Example: For $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$, give S_5 and bound the error.

Step 1. The signs alternate and $1/n$ decreases to zero, so the alternating series test applies and its remainder bound is available.

Step 2. $S_5 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5}$, and the bound is the next term $b_6 = \frac{1}{6}$.

$S_5 = \mathbf{\frac{47}{60}}$ and $|S - S_5| \le \mathbf{\frac{1}{6}}$

Try it: For $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$, give S_4 and the error bound.

check: $S_4 = \mathbf{\frac{115}{144}}$, $|S - S_4| \le \mathbf{\frac{1}{25}}$